Threat Model and Security Control Mapping

Privacy-preserving federated learning for healthcare

PANEL 1 OF 2

01 Adversary (SOURCE)	02 Threat (SCENARIO)	03 Control (MITIGATION)
HC Honest-but-curious aggregation server	T1 Gradient reconstruction	M1 CKKS encryption + aggregate-only threshold decryption
HC Honest-but-curious aggregation server	T2 Membership and property inference	M2 Gradient clipping + DP accountant
HC Honest-but-curious aggregation server	T9 Blockchain metadata leakage	M9 Permissioned ledger, pseudonymous IDs and hash-only on-chain records
MC Malicious or compromised hospital client	T3 Model poisoning and backdoor	M3 Bound proof, reputation weighting, aggregate validation and rollback
MC Malicious or compromised hospital client	T4 Sybil and reputation manipulation	M4 Consortium CA, unique identities and admission control
MC Malicious or compromised hospital client	T7 Client dropout and availability failure	M7 Quorum, timeout and dropout-tolerant threshold

Reading direction: adversary or failure source > threat scenario > primary mitigating control

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PANEL 2 OF 2

01 Adversary (SOURCE)	02 Threat (SCENARIO)	03 Control (MITIGATION)
NET External network attacker	T5 Tampering, MITM and replay	M5 Mutual TLS, signatures, nonce, round ID and timestamp
NET External network attacker	T8 DoS and resource exhaustion	M8 Rate limiting, quotas, circuit breaker and monitoring
COL Colluding participants	T6 Key compromise or collusion	M6 t-of-n key shares, HSM, rotation and revocation
BC Faulty administrator or smart-contract vulnerability	T10 Smart-contract logic failure	M10 Contract audit, multisignature upgrade and emergency pause
BC Faulty administrator or smart-contract vulnerability	T12 Consent violation or revoked-data use	M12 Consent registry, purpose binding, revocation and retention enforcement
USER Malicious inference user	T11 Inference extraction and abuse	M11 Authentication, query limits, minimal output and inference audit

Reading direction: adversary or failure source > threat scenario > primary mitigating control